

Study Notes: Chapter 3 - Describing Bivariate Data

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1 Introduction to Chapter 3

This chapter explores bivariate data—measurements of two variables on a single experimental unit. It covers methods to display and describe relationships between variables, whether categorical or quantitative. Focus is on sections 3.1 (categorical data) and 3.2 (quantitative data), emphasizing graphical and numerical summaries to reveal patterns, such as associations or trends. Detailed Explanation: Bivariate analysis goes beyond univariate by examining how variables interact. For example, does household size affect grocery spending? Graphs and statistics help identify dependencies, strengths, and forms of relationships.

3.1 Describing Bivariate Categorical Data

1.1 Key Concepts

When at least one variable is qualitative (categorical), use pie charts, bar charts (side-by-side or stacked), or line charts to display data. These reveal comparisons across categories or groups.

Definition 1 (Contingency Table). A table showing frequencies (counts) or relative frequencies for combinations of two categorical variables.

Definition 2 (Conditional Distribution). Proportions within one category given a condition on the other (e.g., proportions by rank given college type).

Detailed Explanation: For bivariate categorical data, summarize using frequencies in a contingency table. Visualize with:

Side-by-side bar charts: Compare groups within categories. Stacked bar charts: Show composition within totals. Side-by-side pie charts: Display proportions per group.

Relative frequencies (proportions) aid comparison when group sizes differ.

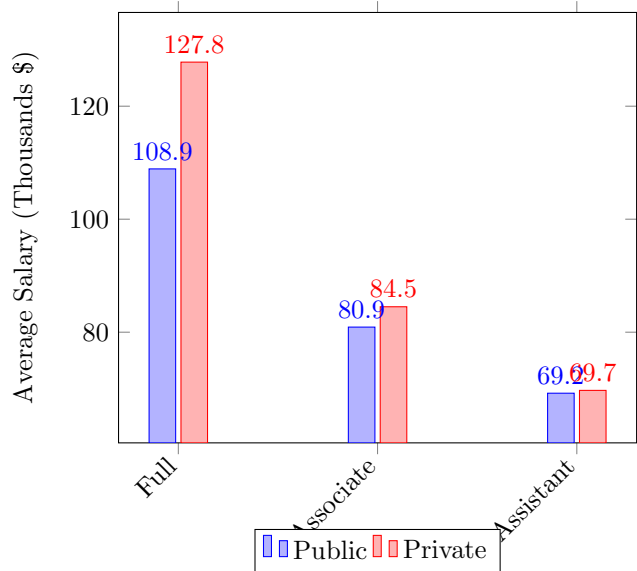
1.2 Examples

Example 1 (Salaries of Professors). Data on 400 professors: rank (Full, Associate, Assistant) and college type (Public, Private). Average salaries (thousands \$):

	Full Professor	Associate Professor	Assistant Professor
Public	108.9	80.9	69.2
Private	127.8	84.5	69.7

Table 1: Salaries of Professors by Rank and Type of College

Side-by-side bar chart (Figure 3.1) shows higher salaries for full professors in private colleges, with smaller



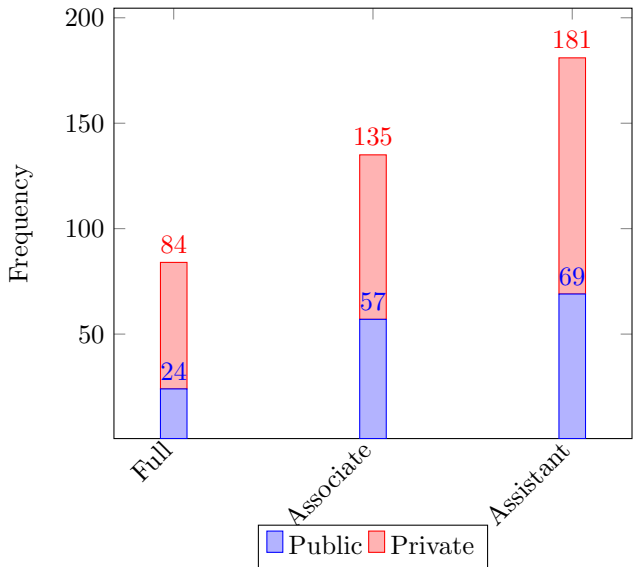
differences at lower ranks.

Example 2 (Number of Professors by Rank and Type). Counts for same categories: Side-by-side pie

	Full Professor	Associate Professor	Assistant Professor	Total
Public	24	57	69	150
Private	60	78	112	250

Table 2: Number of Professors by Rank and Type of College

charts or stacked bar chart show private colleges have more full professors proportionally. Stacked bar chart



visualization:
by type): Public colleges have fewer full professors (16

Conditional distributions (proportions

	Full Professor	Associate Professor	Assistant Professor	Total
Public	0.16	0.38	0.46	1.00
Private	0.24	0.31	0.45	1.00

Table 3: Proportions of Professors by Rank for Public and Private Colleges (Table 3.3)

2 3.2 Describing Bivariate Quantitative Data

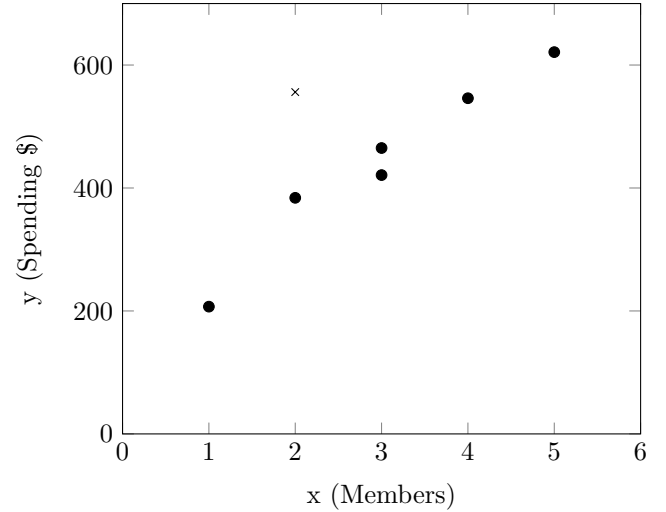
2.1 Scatterplots

Plot quantitative variables on (x, y) axes to visualize patterns. Detailed Explanation: Scatterplots reveal trends (linear, curved, none), strength (tight vs. loose clustering), direction (positive/negative), outliers, or clusters. Independent variable (x) on horizontal, dependent (y) on vertical if applicable.

Example 3 (Household Size and Grocery Spending). Data: x (members), y (spending \$): Scatterplot (Fig-

x	2	3	3	4	1	5
y	384	421	465	546	207	621

Table 4: Household Members and Weekly Grocery Spending



ure 3.4) shows positive linear trend; outlier at (2, 556) deviates.

Example 4 (Wine Price and Demand). Data: price (\$12-16), cases sold (per 10,000 pop.): Scatterplot (Fig-

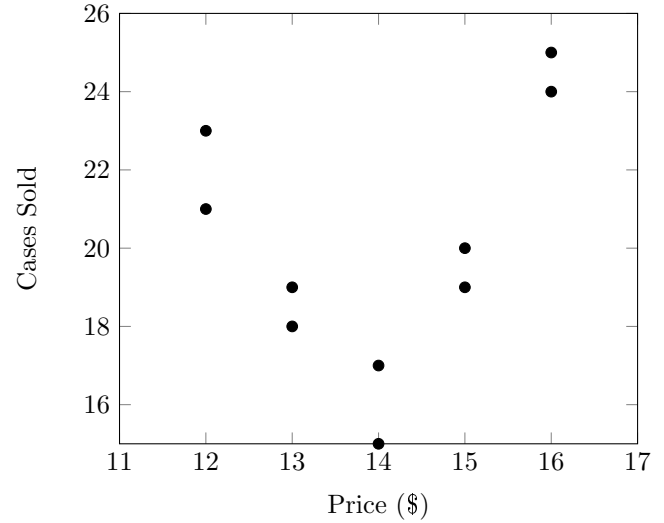


Figure 3.5) shows curved pattern: demand decreases then increases.

2.2 The Correlation Coefficient

Measures strength and direction of linear relationship.

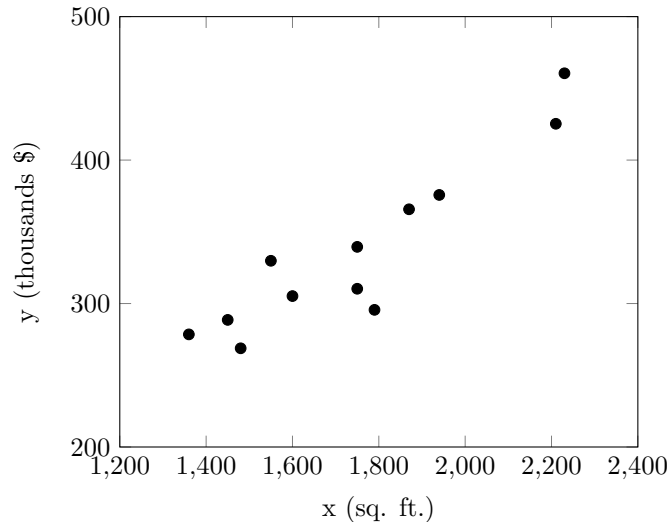
Definition 3 (Sample Correlation Coefficient). $r = \frac{s_{xy}}{s_x s_y}$ where $s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$ (covariance).

Detailed Explanation: r ranges from -1 to 1.

$r \approx 1$: strong positive linear $r \approx -1$: strong negative linear $r \approx 0$: no linear relationship Sign from covariance signs (quadrants in scatterplot).

Computing formula: $s_{xy} = \frac{\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}}{n-1}$

Example 5 (Living Area and Selling Price). Data from Table 3.5 (12 properties, x sq.ft., y thousands).
 $\bar{x} = 1748.333$, $\bar{y} = 336.958$, $s_x = 281.484$, $s_y = 59.759$, $s_{xy} = 15545.197$ $r = \frac{15545.197}{281.484 \times 59.759} \approx 0.924$ Strong



positive linear relationship (Figure 3.6).

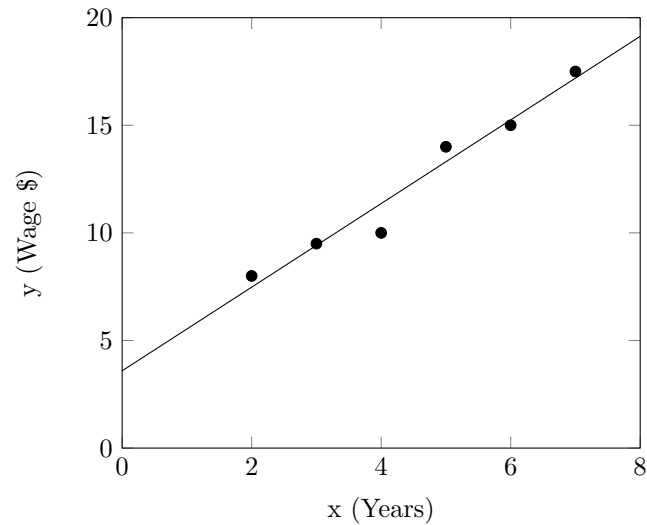
2.3 The Least-Squares Line

Best-fitting line for linear patterns: $\hat{y} = a + bx$

Definition 4 (Slope and Intercept). $b = r \left(\frac{s_y}{s_x} \right)$, $a = \bar{y} - b\bar{x}$

Detailed Explanation: Minimizes squared vertical distances from points to line. Use for prediction when x is independent. b indicates change in y per unit x; a is y when x=0.

Example 6 (Work Experience and Wage). Data: x (years), y (wage \$): $\bar{x} = 4.5$, $\bar{y} = 12.333$, $s_x = 1.871$, $s_y = 3.710$, $r = 0.980$ $b = 0.980 \times \frac{3.710}{1.871} \approx 1.943$, $a = 12.333 - 1.943 \times 4.5 \approx 3.590$ Line: $\hat{y} = 3.590 + 1.943x$



Predict for x=3: $\hat{y} \approx 9.419$

3 Chapter Review

For bivariate data: Use contingency tables and comparative charts for categorical; scatterplots, correlation r (-1 to 1), and least-squares line $\hat{y} = a + bx$ for quantitative. Identify patterns, strength, direction, outliers.